
WaveFlow-UIE: A Single-Pipeline Wavelet-Domain Flow Model for Efficient Underwater Image Restoration

Anonymous Author(s)

Affiliation

Address

email

Abstract

Underwater image restoration remains challenging because color distortion, scattering, and detail loss affect low- and high-frequency image components differently. Recent diffusion-based methods have shown strong restoration quality, but often rely on multi-stage pipelines and costly iterative refinement. We present WaveFlow-UIE, a single-pipeline wavelet-domain flow model that directly learns to transform degraded underwater wavelet representations into clean ones. Given an input image, our method applies a discrete wavelet transform, estimates lightweight physics-inspired degradation cues including transmission and ambient light, and conditions a velocity network trained with flow matching in wavelet space. The training objective combines flow-matching, frequency-weighted reconstruction, perceptual, color-consistency, and physics-consistency losses to jointly preserve global color structure and fine details. Across UIEB, LSUI, UFO-120, and EUVP, WaveFlow-UIE achieves a strong quality-efficiency tradeoff, with especially competitive results on perceptual metrics such as LPIPS and FID. Relative to a strong wavelet-based diffusion baseline, the proposed model substantially reduces inference cost while maintaining or improving restoration quality on several paired benchmarks. Most importantly, we show that reducing the inference process from five flow steps to a single step preserves nearly all of the restoration quality while providing an additional speedup of about $4\times$ on UIEB at 256×256 resolution. This single-step result suggests that wavelet-domain flow matching is an effective alternative to heavier diffusion-style refinement for underwater image restoration.

1 Introduction

Underwater image restoration is challenging because light absorption and scattering cause wavelength-dependent color distortion, haze, low contrast, and loss of fine detail. These degradations reduce both visual quality and the reliability of downstream perception systems used in marine inspection, robotic navigation, and scientific exploration. Recent learning-based methods have improved underwater enhancement substantially, but many still operate purely in the spatial domain and do not explicitly model the different behavior of low- and high-frequency image components.

WF-Diff [Zhao et al., 2024] is a strong frequency-domain restoration framework that addresses this issue by decomposing images into wavelet components and applying a two-stage pipeline: a coarse restoration network followed by diffusion-based frequency residual refinement. While effective, this design increases architectural complexity and inference cost because the refinement stage remains iterative and separate from the main restoration network.

We propose WaveFlow-UIE, a single-pipeline wavelet-domain flow model for underwater image restoration. Instead of predicting a coarse enhanced image and then refining residuals with diffusion,

WaveFlow-UIE directly learns a restoration flow from degraded wavelet representations toward clean wavelet representations. To improve physical plausibility, the model also includes a lightweight physics prior branch that estimates transmission and ambient-light cues and uses them to condition the restoration process. Training combines flow matching, frequency-weighted reconstruction, perceptual supervision, Lab-space color consistency, and physics consistency.

The main empirical result is a favorable quality-efficiency tradeoff relative to the released WF-Diff baseline. In addition, a single-step inference variant preserves nearly all of the full model quality while yielding an additional speedup of about $4\times$ on UIEB at 256×256 resolution, making it one of the strongest practical outcomes of the paper.

Our main contributions are:

- We reformulate wavelet-domain underwater image restoration as a single-pipeline conditional flow problem.
- We introduce physics-prior conditioning through transmission and ambient-light estimation.
- We use frequency-aware, perceptual, and color-consistency supervision to preserve both global color structure and fine details.
- We show a strong speed-quality tradeoff relative to the released WF-Diff baseline, including a strong single-step inference result.
- We show that reducing inference from five flow steps to one preserves nearly all restoration quality while yielding about a $4\times$ speedup on UIEB at 256×256 resolution.

2 Related Work

Underwater image restoration. Underwater enhancement methods aim to recover visually clear images from observations degraded by absorption and scattering. Classical prior-based approaches are interpretable but often brittle under varying water conditions. Learning-based methods such as Water-Net, UWCNN, UIE-WD, and U-shape, learn direct mappings from degraded images to enhanced outputs and generally produce stronger restoration quality, but many operate mainly in RGB image space.

Frequency-domain restoration. Frequency-domain representations are useful in low-level vision because low-frequency components mainly capture color, illumination, and global structure, while high-frequency components capture edges, texture, and fine detail. WF-Diff [Zhao et al., 2024] showed that wavelet and Fourier information can be effectively combined for underwater restoration. Our method keeps the wavelet-domain motivation but replaces the two-stage coarse-plus-diffusion design with a single flow-based pipeline.

Diffusion and flow-based restoration. Diffusion models have achieved strong results in image generation and restoration, but their iterative sampling process can be expensive. Flow matching and rectified flow methods provide an alternative by learning a direct transformation field between source and target representations. WaveFlow-UIE adopts this idea in wavelet space, allowing the model to learn a direct restoration flow without a separate diffusion refinement stage.

3 Method

3.1 Overview

WaveFlow-UIE restores a degraded underwater image by learning a transformation from its degraded wavelet representation to the corresponding clean wavelet representation. Given a degraded image, the model first applies a discrete wavelet transform to separate low- and high-frequency components. A lightweight physics prior branch estimates degradation cues, including transmission and ambient light, and these cues are used to condition a velocity network trained with flow matching. The final restored image is obtained through inverse wavelet reconstruction.

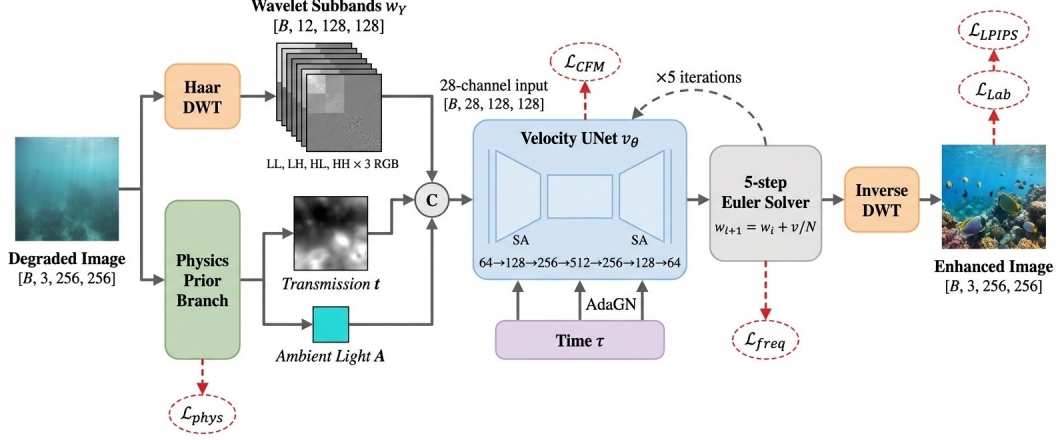


Figure 1: WaveFlow-UIE architecture. A degraded underwater image is decomposed with a Haar discrete wavelet transform, conditioned by a lightweight physics-prior branch that predicts transmission and ambient-light cues, and restored by a velocity U-Net trained with flow matching in wavelet space. The enhanced image is reconstructed with the inverse wavelet transform after iterative flow updates.

3.2 Wavelet-Domain Representation

Underwater degradation affects different visual components differently: color distortion and illumination shifts mainly appear in low-frequency components, while texture loss and edge degradation mainly appear in high-frequency components. We therefore operate in wavelet space instead of directly only in RGB space.

Given a degraded underwater image $x \in \mathbb{R}^{3 \times H \times W}$, we apply the Haar discrete wavelet transform:

$$w_x = \text{DWT}(x), \quad (1)$$

where $w_x \in \mathbb{R}^{12 \times \frac{H}{2} \times \frac{W}{2}}$. Similarly, for the clean reference image y ,

$$w_y = \text{DWT}(y). \quad (2)$$

This representation preserves the information needed for exact reconstruction while separating global structure from fine detail.

3.3 Physics-Prior Conditioning

Underwater image degradation is strongly tied to light propagation in water. Inspired by simplified underwater formation models, we add a lightweight branch that predicts a transmission map t and an ambient-light map A :

$$x \approx y \cdot t + A \cdot (1 - t). \quad (3)$$

These maps are downsampled to wavelet resolution and concatenated into a conditioning tensor:

$$p = [t, A]. \quad (4)$$

This conditioning gives the restoration model explicit information about haze and color cast.

3.4 Wavelet-Domain Flow Matching

Instead of using a separate enhancement stage followed by diffusion refinement, WaveFlow-UIE directly learns a restoration flow in wavelet space. During training, we sample a time value $\tau \sim \mathcal{U}(0, 1)$ and construct an interpolated wavelet state:

$$w_\tau = (1 - \tau)w_x + \tau w_y. \quad (5)$$

The target velocity is the direction from the degraded wavelet representation to the clean wavelet representation:

$$v^* = w_y - w_x. \quad (6)$$

102 A conditional velocity network predicts this direction:

$$\hat{v} = v_{\theta}(w_{\tau}, \tau, w_x, p), \quad (7)$$

103 where w_{τ} is the current wavelet state, τ is the flow time, w_x is the degraded condition, and p is the
104 physics-prior condition.

105 The predicted clean wavelet representation is:

$$\hat{w}_y = w_x + \hat{v}, \quad (8)$$

106 and the restored image is reconstructed with the inverse wavelet transform:

$$\hat{y} = \text{IDWT}(\hat{w}_y). \quad (9)$$

107 3.5 Training Objective

108 The total training loss combines several complementary terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cfm}} \mathcal{L}_{\text{cfm}} + \lambda_{\text{freq}} \mathcal{L}_{\text{freq}} + \lambda_{\text{lpips}} \mathcal{L}_{\text{lpips}} + \lambda_{\text{lab}} \mathcal{L}_{\text{lab}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}. \quad (10)$$

109 Here, $\mathcal{L}_{\text{cfm}}$ is the conditional flow matching loss, $\mathcal{L}_{\text{freq}}$ is a frequency-weighted wavelet reconstruction
110 loss that emphasizes high-frequency detail, $\mathcal{L}_{\text{lpips}}$ is the perceptual loss, $\mathcal{L}_{\text{lab}}$ enforces Lab-space
111 color consistency, and $\mathcal{L}_{\text{phys}}$ supervises the physics-prior branch. This combination is intended to
112 preserve both color fidelity and fine detail while keeping the model physically grounded.

113 3.6 Inference

114 At inference time, the model starts from the degraded wavelet representation and updates it with the
115 predicted velocity field:

$$w_{k+1} = w_k + \Delta t \cdot v_{\theta}(w_k, t_k, w_x, p), \quad (11)$$

116 where k denotes the inference step and Δt is the step size. In the default setting, we use $K = 5$ Euler
117 steps. We also evaluate a single-step variant ($K = 1$) to study the quality-efficiency tradeoff.

118 4 Experiments

119 4.1 Setup

120 **Implementation details.** The model is implemented in PyTorch. We use AdamW with $\beta_1 = 0.9$,
121 $\beta_2 = 0.999$, weight decay 1×10^{-4} , an initial learning rate of 2×10^{-4} , linear warm-up for the
122 first 5,000 iterations, and cosine decay to 1×10^{-6} . Training uses 256×256 patches, an effective
123 batch size of 8, mixed precision, and gradient clipping at 1.0. The default inference setting uses
124 $K = 5$ Euler steps with step size $\Delta t = 0.2$. The main WaveFlow models were trained on a single
125 NVIDIA GeForce RTX 5080 GPU, and all reproduced evaluation runs were measured on a single
126 NVIDIA GeForce RTX 5070 Ti GPU with 16 GB VRAM. The three main WaveFlow checkpoints
127 correspond to approximately 200k iterations on UIEB and 2M iterations on each of LSUI and the
128 mixed UIEB+LSUI training setup.

129 **Datasets and metrics.** We train on UIEB [Li et al., 2020] and LSUI [Peng et al., 2023], and evaluate
130 on UIEB, LSUI, UFO-120 [Islam et al., 2020a], EUVP [Islam et al., 2020b], U45 [Li et al., 2019],
131 and C60. For paired benchmarks we report PSNR $\uparrow$, SSIM $\uparrow$, LPIPS $\downarrow$, and FID $\downarrow$. For non-reference
132 benchmarks we additionally report UCIQE $\uparrow$ and UIQM $\uparrow$.

133 **Comparison methods.** We compare WaveFlow-UIE against reproduced public checkpoints for
134 WF-Diff, UIE-WD, UWCNN, Water-Net, and U-shape in the main quantitative benchmark. UIEC²-
135 Net and DM-water are discussed as relevant prior underwater enhancement methods, but the final
136 main-paper quantitative tables focus on the reproduced baseline subset above to keep the comparison
137 consistent across datasets and within the page budget. To avoid overclaiming exact protocol repro-
138 duction across papers, all reproduced models are evaluated under a unified local benchmark protocol
139 with consistent preprocessing, metric computation, and runtime measurement on the same hardware.

Table 1: Quantitative comparison on the public UIEB and LSUI splits at standardized 256×256 resolution. Best results are highlighted in bold and second-best results are underlined. The Ours row uses WaveFlow-UIEB on UIEB and WaveFlow-LSUI on LSUI.

Method	UIEB						LSUI					
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$
UIEWD	14.8806	0.7387	0.3771	92.3536	0.5704	4.0653	17.8399	0.8051	0.3348	53.1968	0.5642	4.4957
UWCNN	19.5062	0.8482	0.3080	72.2604	0.5451	3.8867	21.0453	0.8636	0.3264	54.8434	0.5560	4.0106
WaterNet	23.6456	0.9106	<u>0.1623</u>	<u>43.7351</u>	0.5826	4.2446	24.9394	0.9015	0.2227	39.2744	<u>0.5857</u>	<u>4.4053</u>
Ushape	24.6389	0.8479	0.1940	48.6329	0.5857	4.2037	29.1284	<u>0.9172</u>	<u>0.1449</u>	22.9495	0.5865	4.3093
WF-Diff	<u>24.7457</u>	<u>0.9150</u>	0.1692	53.9730	<u>0.5847</u>	4.1871	28.2709	0.9112	0.1506	33.2456	0.5734	4.3176
Ours	24.9818	0.9316	0.1390	29.7382	0.5835	4.0450	<u>28.4551</u>	0.9246	0.1216	<u>25.9970</u>	0.5743	4.2832

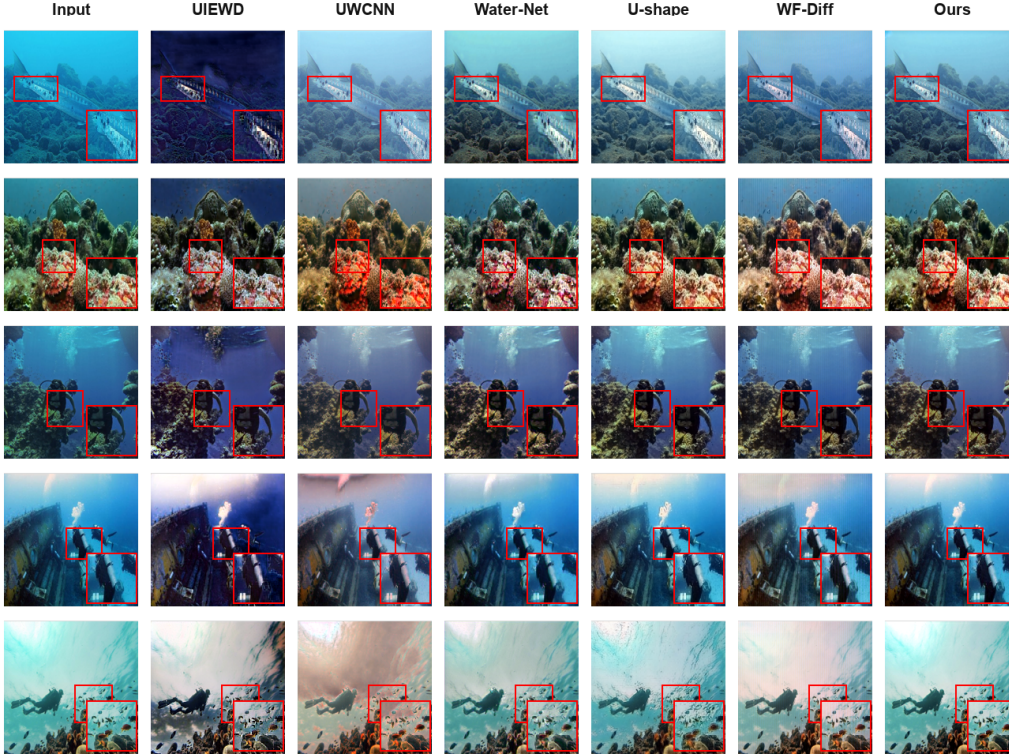


Figure 2: Representative qualitative comparison on five UIEB test examples. To keep the qualitative study aligned with the main quantitative tables, we compare against the same baseline subset: UIEWD, UWCNN, Water-Net, U-shape, and WF-Diff. WaveFlow-UIE typically restores stronger color contrast than UIE-WD and UWCNN while preserving fine structure more consistently than the heavier WF-Diff baseline.

4.2 Main Quantitative Results

Table 1 shows that WaveFlow-UIE is especially strong on perceptual metrics at standardized resolution. On UIEB, our method achieves the best LPIPS and the second-best PSNR, SSIM, and FID, indicating that it produces visually closer reconstructions while remaining competitive in pixel-level fidelity. On LSUI, WaveFlow-UIE obtains the best SSIM and LPIPS and the second-best PSNR and FID. These results suggest that the proposed wavelet-domain flow formulation is effective at preserving structure and perceptual detail across the two main paired benchmarks.

Figure 2 provides a qualitative view of the same comparison. Across these examples, WaveFlow-UIE tends to recover more balanced color and clearer local detail than the weaker feed-forward baselines, while remaining visually competitive with the stronger wavelet-based methods. This qualitative

Table 2: Benchmark results on UFO-120, EUVP, U45, and C60 at 256×256 .

Method	UFO-120						EUVP						U45		C60	
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$	UCIQE $\uparrow$	UIQM $\uparrow$	UCIQE $\uparrow$	UIQM $\uparrow$
UIEWD	19.4465	0.8122	0.3237	64.2594	0.5925	4.4257	19.3659	0.7946	0.3412	46.1214	0.5797	4.4398	0.5647	3.9488	0.5227	4.0000
UWCNN	20.7244	0.8625	0.2909	74.2507	0.5905	4.0840	21.5804	0.8521	0.3238	51.8359	0.5778	4.1469	0.5287	3.8195	0.5210	3.5638
WaterNet	24.2772	0.8780	0.2591	52.9071	0.6010	4.3494	24.4686	0.8644	0.2876	39.9792	0.5931	4.4131	0.5755	4.2914	0.5637	4.0146
Ushape	29.5540	<u>0.9111</u>	0.1792	38.8640	<u>0.5997</u>	4.2437	31.4794	0.9030	0.1634	22.9462	<u>0.5886</u>	4.2795	0.5860	<u>4.2262</u>	<u>0.5585</u>	3.8208
WF-Diff	30.6599	0.9044	<u>0.1336</u>	<u>36.0713</u>	0.5877	4.2971	<u>29.9486</u>	<u>0.8945</u>	<u>0.1589</u>	<u>23.0850</u>	0.5800	4.3459	–	–	–	–
Ours	<u>30.3101</u>	0.9146	0.1138	33.7577	0.5904	4.2741	29.2063	0.8931	0.1503	26.3554	0.5826	4.3222	0.5626	4.0394	0.5501	3.7271

Table 3: Inference time comparison on the public UIEB split. Lower is better. The Ours row uses WaveFlow-UIEB on UIEB.

Method	UIEB	
	Native Time $\downarrow$	256 Time $\downarrow$
UIEWD	24.28	3.32
UWCNN	141.12	12.50
WaterNet	118.50	4.68
Ushape	–	28.01
WF-Diff	3392.32	301.30
Ours	444.83	79.31

behavior is consistent with the quantitative pattern in Table 1, where the model is strongest on perceptual fidelity.

Table 2 summarizes cross-dataset evaluation on UFO-120, EUVP, U45, and C60 at standardized 256×256 resolution. Using the LSUI-trained checkpoint, WaveFlow-UIE generalizes strongly on the paired benchmarks: on UFO-120 it achieves the best SSIM, LPIPS, and FID while remaining second-best in PSNR, indicating strong preservation of structure and perceptual quality under domain shift. On EUVP, our method obtains the best LPIPS, showing that its outputs remain perceptually close to the reference images, although U-shape and WF-Diff are stronger on PSNR, SSIM, and FID. On the non-reference sets U45 and C60, the mixed-data WaveFlow-BOTH checkpoint remains competitive but does not lead the UIQM/UCIQE rankings, suggesting that WaveFlow generalizes more reliably on paired cross-dataset benchmarks than on fully reference-free challenge images.

4.3 Efficiency, Single-Step Inference, and Ablation

Table 3 compares inference time on UIEB. WaveFlow-UIE is substantially faster than WF-Diff, reducing native-resolution runtime from 3392.32 ms to 444.83 ms and the 256×256 runtime from 301.30 ms to 79.31 ms. This supports the motivation for replacing the two-stage WF-Diff pipeline with a single flow-based pipeline. However, WaveFlow-UIE remains slower than lightweight feed-forward methods such as UIE-WD and Water-Net because it still performs multiple flow update steps during inference.

Table 4 reports the ablation study on UIEB at standardized 256×256 resolution. We compare the full model against variants that remove the physics prior, remove the frequency weighting, remove the LPIPS loss, remove the Lab color loss, and reduce the inference process to a single flow step ($K = 1$). Since the goal is to measure relative component importance under a consistent local benchmark, we focus on a single representative paired dataset and report fair timing only for the full model and the single-step variant.

The full WaveFlow-UIE model achieves the best overall balance across PSNR, SSIM, LPIPS, and FID, showing that the final design is not driven by any single component alone. The most practically important ablation is the single-step setting: reducing inference from five flow steps to one changes PSNR only from 24.98 to 24.94, SSIM only from 0.9316 to 0.9306, LPIPS only from 0.1390 to 0.1403, and FID only from 29.74 to 30.66, while reducing runtime from 79.31 ms to 19.55 ms. This shows that most of the restoration performance can be retained with a much cheaper inference process. Removing the physics prior reduces performance across all main restoration metrics, indicating that explicit transmission and ambient-light conditioning helps the model correct underwater degradation more consistently. Removing the frequency weighting also hurts performance, especially in PSNR and

Table 4: UIEB ablation results at standardized 256×256 resolution.

Configuration	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$	Time $\downarrow$ (ms)
Full model ($K=5$)	24.9818	0.9316	0.1390	29.7382	0.5835	4.0450	79.31
$K=1$ single-step	24.9382	0.9306	0.1403	30.6582	0.5827	4.0638	19.55
No physics prior	24.5000	0.9234	0.1469	35.7000	0.5853	4.0912	–
No frequency weighting	24.0902	0.9298	0.1426	33.8364	0.5852	4.0270	–
No LPIPS loss	24.0288	0.9047	0.1574	35.1146	0.5753	4.2561	–
No Lab color loss	24.1482	0.9220	0.1848	44.0632	0.5819	4.0640	–

FID, which suggests that emphasizing high-frequency wavelet errors is important for preserving detail while maintaining global fidelity. The no-LPIPS and no-Lab variants cause the largest perceptual degradation, with the no-Lab model producing the worst LPIPS and FID among the trained loss ablations.

5 Limitations and Broader Impact

WaveFlow-UIE is strongest on paired restoration benchmarks, where it consistently achieves strong perceptual quality and a favorable runtime-quality tradeoff relative to the released WF-Diff baseline. However, performance on fully reference-free challenge sets remains more mixed, and the model is still slower than lightweight feed-forward baselines because inference remains iterative. The current study also relies on a reproduced public-checkpoint benchmark rather than an exact reimplementation of every original paper protocol, so our claims should be interpreted within that unified local evaluation setting.

The method has potentially positive impact for underwater inspection, marine exploration, and robotic perception by improving visibility in degraded underwater imagery. At the same time, restoration errors may alter color or structure in ways that could affect downstream decisions if the outputs are used without validation in safety-critical settings. For that reason, the method should be treated as a perceptual restoration tool rather than a substitute for task-specific robustness evaluation.

6 Conclusion

We presented WaveFlow-UIE, a single-pipeline wavelet-domain flow model for underwater image restoration. The method keeps the key frequency-domain insight of wavelet-based restoration while removing the need for a separate diffusion-style refinement stage. Across paired and cross-dataset benchmarks, WaveFlow-UIE shows a strong quality-efficiency tradeoff relative to the released WF-Diff baseline and particularly strong performance on perceptual metrics. The strongest practical result is that a single-step variant retains nearly all of the five-step model quality while delivering about a $4\times$ speedup on UIEB at 256×256 resolution. The strongest evidence in the current benchmark is on paired restoration tasks; performance on fully reference-free challenge sets is more mixed and remains a useful direction for future work.

References

- Chongyi Li, Chunle Guo, Wenqi Ren, Runmin Cong, Junhui Hou, Sam Kwong, and Dacheng Tao. An underwater image enhancement benchmark dataset and beyond. *IEEE Transactions on Image Processing*, 29:4376–4389, 2020.
- L. Peng, C. Zhu, and L. Bian. U-shape Transformer for underwater image enhancement. *IEEE Transactions on Image Processing*, 32:3066–3079, 2023.
- Md Jahidul Islam, Peigen Luo, and Junaed Sattar. Simultaneous enhancement and super-resolution of underwater imagery for improved visual perception. In *Robotics: Science and Systems*, 2020.
- Md Jahidul Islam, Youya Xia, and Junaed Sattar. Fast underwater image enhancement for improved visual perception. *IEEE Robotics and Automation Letters*, 5(2):3227–3234, 2020.
- H. Li, J. Li, and W. Wang. A fusion adversarial underwater image enhancement network with a public test dataset. *arXiv preprint arXiv:1906.06819*, 2019.

- 222 Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment:
223 From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–
224 612, 2004.
- 225 Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable
226 effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on*
227 *Computer Vision and Pattern Recognition*, pages 586–595, 2018.
- 228 Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter.
229 GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Advances*
230 *in Neural Information Processing Systems*, 2017.
- 231 Karen Panetta, Chen Gao, and Sos Agaian. Human-visual-system-inspired underwater image quality
232 measures. *IEEE Journal of Oceanic Engineering*, 41(3):541–551, 2015.
- 233 M. Yang and A. Sowmya. An underwater color image quality evaluation metric. *IEEE Transactions*
234 *on Image Processing*, 24(12):6062–6071, 2015.
- 235 Chen Zhao, Weiling Cai, Chenyu Dong, and Chengwei Hu. Wavelet-based Fourier information
236 interaction with frequency diffusion adjustment for underwater image restoration. In *Proceedings*
237 *of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8281–8291,
238 2024.

A Additional Results

A.1 Native UIEB and LSUI

Table 5: Quantitative comparison on the public UIEB and LSUI splits at native resolution. Best results are highlighted in bold and second-best results are underlined. The Ours row uses WaveFlow-UIEB on UIEB and WaveFlow-LSUI on LSUI. Methods without valid native reproduced rows in the current benchmark setup are omitted.

Method	UIEB						LSUI					
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$
UIEWD	14.3053	0.7507	0.3945	97.1222	0.5622	3.8048	17.4945	0.7853	0.3582	55.0274	0.5600	4.2795
UWCNN	19.1571	0.8601	0.3312	74.7177	0.5378	3.3864	20.6892	0.8354	0.3660	54.3170	0.5530	3.7243
WaterNet	23.1406	0.9156	<u>0.1833</u>	<u>36.1890</u>	0.5806	3.8336	24.4207	0.8768	0.2558	38.1637	0.5856	<u>4.2284</u>
WF-Diff	<u>23.6565</u>	<u>0.9346</u>	0.1963	38.5893	0.5831	<u>3.8225</u>	<u>25.9974</u>	0.8973	<u>0.1957</u>	<u>27.5106</u>	0.5699	4.0466
Ours	24.6786	0.9350	0.1589	27.6999	<u>0.5820</u>	3.6330	27.2355	<u>0.8949</u>	0.1519	27.0609	<u>0.5728</u>	4.1128

A.2 Native UFO-120, EUVP, U45, and C60

Table 6: Benchmark results on UFO-120, EUVP, U45, and C60 at native resolution. Best results are highlighted in bold and second-best results are underlined. The Ours row uses WaveFlow-LSUI on UFO-120/EUVP and WaveFlow-BOTH on U45/C60. The WF-Diff row uses WF-Diff-LSUI on UFO-120/EUVP and a reproduced native U45 run; no valid native C60 row is available for WF-Diff in the current benchmark setup.

Method	UFO-120						EUVP						U45		C60	
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$	UCIQE $\uparrow$	UIQM $\uparrow$	UCIQE $\uparrow$	UIQM $\uparrow$
UIEWD	19.1212	0.7679	0.3678	66.2856	<u>0.5909</u>	4.3050	19.3659	0.7946	0.3412	46.1214	0.5797	4.4398	0.5647	3.9488	0.5042	2.9611
UWCNN	20.3827	0.8070	0.3515	75.4400	0.5885	3.9580	21.5804	0.8521	0.3238	51.8359	0.5778	4.1469	0.5287	3.8195	0.5073	2.1880
WaterNet	23.6631	0.8291	0.3083	57.3224	0.6008	4.2826	24.4686	0.8644	0.2876	39.9792	0.5931	<u>4.4131</u>	<u>0.5755</u>	4.2914	0.5592	<u>2.6711</u>
WF-Diff	28.8062	0.8716	<u>0.1707</u>	34.1463	0.5856	4.1173	29.9486	0.8945	<u>0.1589</u>	23.0850	0.5800	4.3459	0.5819	<u>4.2097</u>	–	–
Ours	<u>28.2632</u>	<u>0.8561</u>	0.1616	<u>40.2139</u>	<u>0.5909</u>	4.2255	<u>29.2063</u>	<u>0.8931</u>	0.1503	<u>26.3554</u>	<u>0.5826</u>	4.3222	0.5626	4.0394	<u>0.5270</u>	2.6034

A.3 Native Ablation Results

Table 7: UIEB ablation results at native resolution.

Configuration	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	UCIQE $\uparrow$	UIQM $\uparrow$	Time $\downarrow$ (ms)
Full model ($K=5$)	24.6786	0.9350	0.1589	27.6999	0.5820	3.6330	444.83
$K=1$ single-step	24.7144	0.9344	0.1591	27.7457	0.5798	3.6600	117.43
No physics prior	24.1800	0.9329	0.1603	30.6200	0.5778	3.6756	–
No frequency weighting	23.7948	0.9330	0.1606	31.1261	0.5812	3.6619	–
No LPIPS loss	24.0765	0.9205	0.1796	30.8640	0.5750	3.7592	–
No Lab color loss	23.8283	0.9261	0.1980	39.1097	0.5789	3.6965	–

B NeurIPS Paper Checklist

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and introduction state the main claim that WaveFlow-UIE replaces a heavier diffusion-style refinement stage with a single-pipeline wavelet-domain flow model and achieves a favorable quality-efficiency tradeoff relative to the released WF-Diff baseline; see the Abstract, Section 1, and Section 5.

2. Limitations

254 Question: Does the paper discuss the limitations of the work performed by the authors?
 255 Answer: [Yes]
 256 Justification: The paper now includes an explicit “Limitations and Broader Impact” section
 257 stating that the strongest evidence is on paired restoration tasks, that performance on
 258 fully reference-free challenge sets is more mixed, and that the model remains slower than
 259 lightweight feed-forward baselines; see Section 5.

260 **3. Theory assumptions and proofs**
 261 Question: For each theoretical result, does the paper provide the full set of assumptions and
 262 a complete (and correct) proof?
 263 Answer: [N/A]
 264 Justification: The paper is primarily empirical and does not present new theorems or formal
 265 proofs.

266 **4. Experimental result reproducibility**
 267 Question: Does the paper fully disclose all the information needed to reproduce the main ex-
 268 perimental results of the paper to the extent that it affects the main claims and/or conclusions
 269 of the paper (regardless of whether the code and data are provided or not)?
 270 Answer: [Yes]
 271 Justification: The paper describes the model design, losses, inference procedure, datasets,
 272 metrics, and main training and evaluation settings, and it states that reproduced public
 273 checkpoints are evaluated under a unified local benchmark protocol; see Sections 3 and 4
 274 and the appendix.

275 **5. Open access to data and code**
 276 Question: Does the paper provide open access to the data and code, with sufficient instruc-
 277 tions to faithfully reproduce the main experimental results, as described in supplemental
 278 material?
 279 Answer: [No]
 280 Justification: The current submission does not provide anonymized public code or a public
 281 release package in the paper or appendix. The paper does provide methodological and
 282 experimental detail, but open-access code and release instructions are not yet part of the
 283 submission.

284 **6. Experimental setting/details**
 285 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
 286 rameters, how they were chosen, type of optimizer) necessary to understand the results?
 287 Answer: [Yes]
 288 Justification: The paper reports datasets, metrics, the optimizer, learning rate schedule, patch
 289 size, effective batch size, inference steps, hardware context, and the main loss design in
 290 Sections 3 and 4, with additional results tables in the appendix.

291 **7. Experiment statistical significance**
 292 Question: Does the paper report error bars suitably and correctly defined or other appropriate
 293 information about the statistical significance of the experiments?
 294 Answer: [No]
 295 Justification: The current paper reports average benchmark metrics but does not include
 296 repeated-seed variance, confidence intervals, or statistical significance tests.

297 **8. Experiments compute resources**
 298 Question: For each experiment, does the paper provide sufficient information on the com-
 299 puter resources (type of compute workers, memory, time of execution) needed to reproduce
 300 the experiments?
 301 Answer: [Yes]
 302 Justification: The paper now reports the training GPU, evaluation GPU, available VRAM for
 303 the evaluation machine, and the scale of the main training runs in terms of iteration counts,
 304 together with explicit runtime tables for inference; see Section 4.

305 **9. Code of ethics**

306 Question: Does the research conducted in the paper conform, in every respect, with the

307 NeurIPS Code of Ethics?

308 Answer: [\[Yes\]](#)

309 Justification: The work is a low-level image restoration method evaluated on public datasets

310 and reproduced public checkpoints, and the submission does not involve deceptive evaluation

311 practices, human-subject experiments, or undisclosed conflicts with the NeurIPS Code of

312 Ethics.

313 **10. Broader impacts**

314 Question: Does the paper discuss both potential positive societal impacts and negative

315 societal impacts of the work performed?

316 Answer: [\[Yes\]](#)

317 Justification: The paper now includes a “Limitations and Broader Impact” section that

318 discusses positive uses in underwater inspection and exploration and the risk that restoration

319 artifacts could affect downstream decisions in safety-critical settings; see Section 5.

320 **11. Safeguards**

321 Question: Does the paper describe safeguards that have been put in place for responsible

322 release of data or models that have a high risk for misuse?

323 Answer: [\[N/A\]](#)

324 Justification: The work does not introduce a high-risk generative model or scraped dataset

325 release, and the current submission does not release a potentially dual-use model artifact.

326 **12. Licenses for existing assets**

327 Question: Are the creators or original owners of assets used in the paper properly credited

328 and are the license and terms of use explicitly mentioned and properly respected?

329 Answer: [\[No\]](#)

330 Justification: The paper cites the original datasets and baseline methods, but it does not yet

331 explicitly list the licenses and terms of use for all datasets, checkpoints, and external code

332 assets used in the reproduced benchmark.

333 **13. New assets**

334 Question: Are new assets introduced in the paper well documented and is the documentation

335 provided alongside the assets?

336 Answer: [\[N/A\]](#)

337 Justification: The current submission does not publicly release a new dataset, code package,

338 or model asset as part of the anonymous submission.

339 **14. Crowdsourcing and research with human subjects**

340 Question: For crowdsourcing experiments and research with human subjects, does the paper

341 include the full text of instructions given to participants and screenshots, if applicable, as

342 well as details about compensation (if any)?

343 Answer: [\[N/A\]](#)

344 Justification: The paper does not involve crowdsourcing experiments or research with human

345 subjects.

346 **15. Institutional review board (IRB) approvals or equivalent for research with human**

347 **subjects**

348 Question: Does the paper describe potential risks incurred by study participants, whether

349 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

350 approvals (or an equivalent approval/review based on the requirements of your country or

351 institution) were obtained?

352 Answer: [\[N/A\]](#)

353 Justification: The paper does not involve crowdsourcing nor research with human subjects.

354 **16. Declaration of LLM usage**
355 Question: Does the paper describe the usage of LLMs if it is an important, original, or
356 non-standard component of the core methods in this research?
357 Answer: [N/A]
358 Justification: LLMs are not an important, original, or non-standard component of the
359 proposed method or experiments. Any LLM use, if any, was limited to writing or formatting
360 assistance rather than the core methodology.